Lab 1 – Take the Lead Product Description

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CS 411W

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1 Introduction

A sick or injured pet often needs care before its owner has time to plan for the cost. A 2025 Gallup study found that 52% of U.S. pet owners skipped or declined veterinary care their pet needed during the past year, and 71% of those owners named cost as the main reason (Hrynowski & Lall, 2025). The same study found that about two-thirds of owners could pay $1,000 or less for lifesaving treatment.

Veterinarians see the same problem from the other side of the exam table. In a 2026 Gallup survey, 94% of veterinarians said a client's finances often or sometimes stop them from giving the recommended treatment (Kemp, 2026). A pet whose care is delayed or refused can get worse, and the owner, the veterinarian, and the animal all pay the price.

An effective solution must let treatment begin before the full bill is paid, give the owner a realistic way to raise the money, and give the hospital confidence that the bill will be covered. Take the Lead is a website application built around these three needs. Take the Lead guarantees payment to the veterinary hospital so treatment can start, and the owner raises the money through a 30-day online fundraiser shared with friends, family, and the community.

2 Product Description

Take the Lead connects three groups: pet owners who need help with a veterinary bill, the veterinary hospital that treats the pet, and the people who want to help. The process has three steps. The owner signs up as a Member and asks Take the Lead to back the bill. Take the Lead sends the hospital either a Promise to Pay or a Minimum Down Payment so treatment can begin. The final step is a Story, a fundraising page that the Member shares online for 30 days.

Donations collected by the end of the Story first cover any amount Take the Lead paid up front and then the rest of the hospital bill. Leftover money belongs to the Member, and any unpaid balance remains the Member's responsibility. The goal of Take the Lead is to keep cost from being the reason a pet goes without needed care. Take the Lead will reach that goal through a payment guarantee for hospitals, a simple fundraising tool for owners, and plain-language guides on veterinary costs and pet insurance.

2.1 Key Product Features and Capabilities

Take the Lead has three main capabilities. The payment guarantee is what sets Take the Lead apart. Crowdfunding websites release money only after donors give it, which can be too late for a pet in an emergency. Take the Lead backs the bill first, through a Promise to Pay or a Minimum Down Payment, so the hospital can treat the pet right away.

The Story is the fundraising tool. The Member has 48 hours after the visit to set up a Story, which shows a short description of the pet, the amount needed, and progress toward that amount. Anyone with the link can donate while the Story is open.

The resource pages help owners prepare for the next bill. These pages include cost guides for common treatments, basic information on pet insurance, and links to support for veterinary staff dealing with stress and burnout.

These three capabilities address both sides of the problem. The owner gets time and help to pay, and the hospital gets paid for the care it provides.

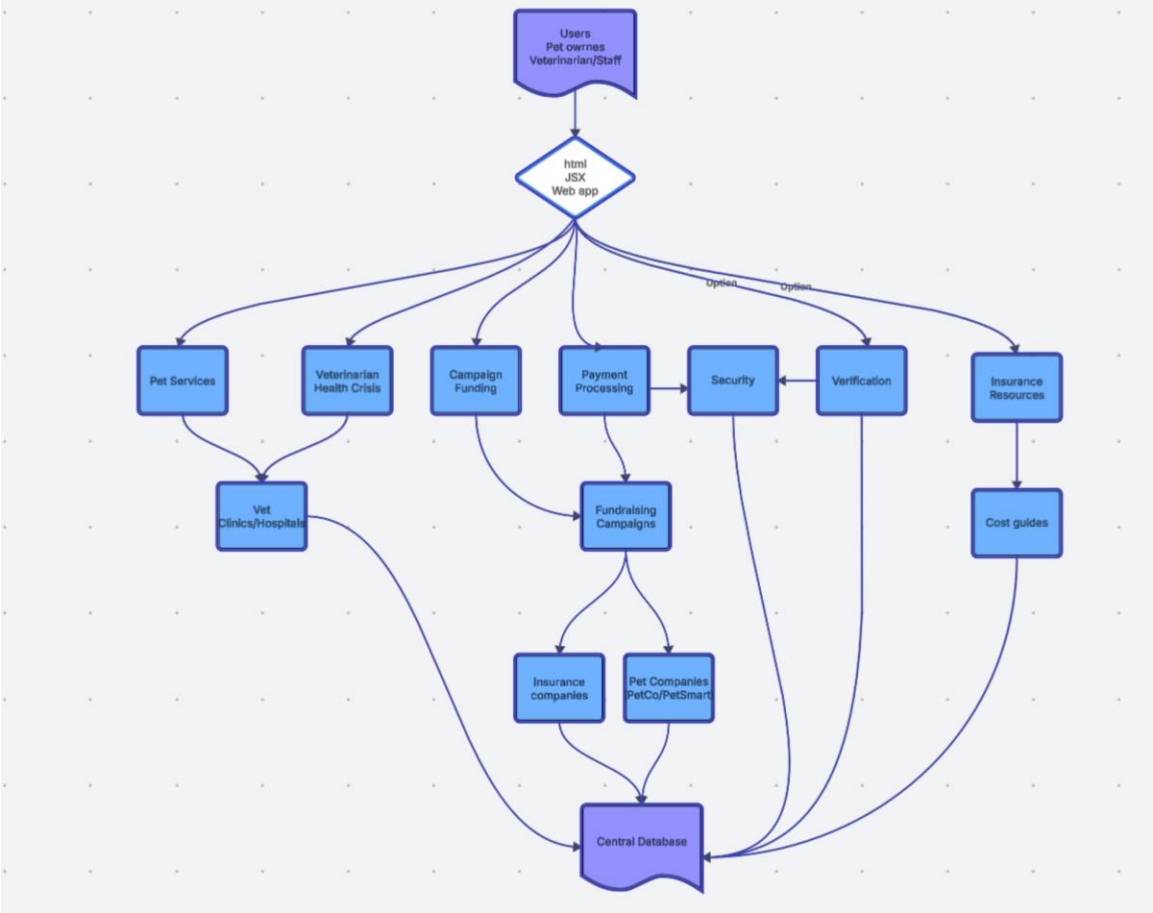
2.2 Major Components (Hardware/Software)

Take the Lead runs entirely in a web browser, so the team will not build any physical devices. A pet owner, donor, or hospital staff member needs only a phone, tablet, or computer with internet access. The website application itself needs hosting and data storage, and the team is considering Cloudflare, which provides both as online services.

SvelteKit is the main framework and handles both the pages users see and the logic that runs on the server. Tailwind CSS sets the look of each page, and shadcn and Bits UI provide ready-made buttons, menus, and forms. Member accounts, Stories, and donations are saved in an SQLite database, and Prisma ORM lets the code read and write that data without hand-written database queries. The better-auth library handles authentication so each Member account stays private.

The CS 410 Major Functional Component Diagram (MFCD) in Figure 1 shows how the parts of Take the Lead fit together.

Figure 1: Take the Lead Major Functional Component Diagram



Every user enters through the website application at the top of the diagram. The paths then split by purpose. Campaign and payment components handle fundraising, security and verification components protect accounts and payments, and resource components provide cost guides, insurance information, and links to veterinary hospitals. All paths end at a central database that stores the information for the whole system.

3 Case Study

The case study shows how Take the Lead would work for a small beta test group before it is offered to the public. The group is one small-animal veterinary hospital in Norfolk, Virginia, five pet-owner households who are clients of that hospital, and the friends and family who donate to their Stories. The group will use the Take the Lead prototype for 30 days, the length of one Story, and report what worked and what did not. The case study connects the real-world product described in Sections 1 and 2 to the description of the prototype that will follow.

3.1 Case Study Participants and Roles

Each participant in the case study plays a different part in the process. The pet owners request help and run the Stories, the hospital staff check each request and treat the pets, and the donors give money and share the Story links. A Take the Lead administrator reviews each request and watches the totals so the case study runs smoothly. Table 1 lists each participant, the role each one plays, and what each one gains.

Table 1: Case Study Participants and Roles

Participant	Role in the Case Study	What Take the Lead Yields
Pet owners (five households)	Sign up as Members, request a payment guarantee, and create and share a Story	Treatment that starts right away and 30 days to raise the money
Veterinary hospital staff	Confirm the bill amount and accept the Promise to Pay or Minimum Down Payment	Payment backed by Take the Lead and fewer declined treatments
Donors	Open a shared Story link and give a donation	A simple and direct way to help a specific pet
Take the Lead administrator	Approve requests and track each Story until it closes	Feedback and data to improve the product before launch

3.2 Features Demonstrated

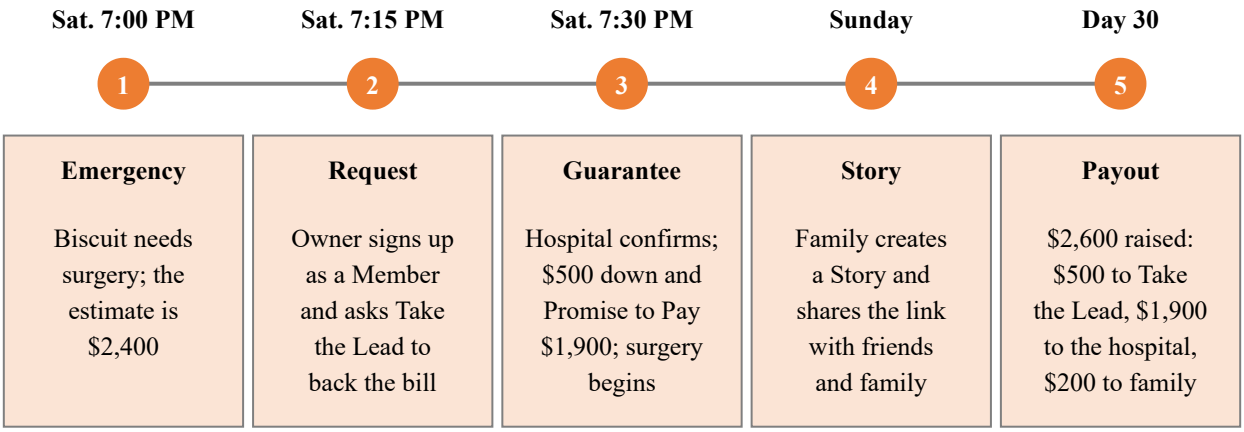
The case study exercises every step of the Take the Lead process. Pet owners create Member accounts that are protected by a sign-up and login. Each Member enters the pet's information and the hospital estimate and then asks Take the Lead to back the bill. Hospital staff use their own account to confirm the estimate and receive either a Promise to Pay or a Minimum Down Payment. These features yield the most important result for the participants, which is that the pet is treated the same day instead of waiting for money.

After the visit, each Member has 48 hours to set up a Story with a photo, a short description of the pet, and the amount needed. Donors can give through the shared link, and a progress bar shows how close the Story is to its goal. When the Story closes, Take the Lead applies the donations in a set order. Donations first repay any amount Take the Lead paid up front, then cover the rest of the hospital bill, and any leftover money goes to the Member. The resource pages are also part of the case study, so Members can read the cost guides and pet insurance information while their Story is open.

3.3 Use Case Scenario: A Day in the Life

One household in the beta group is a family with a five-year-old dog named Biscuit. On a Saturday evening Biscuit swallows a sock and starts vomiting, and the family rushes Biscuit to the partner hospital. The veterinarian recommends surgery that will cost $2,400. The family can afford groceries and rent that month, but they cannot pay $2,400 that night. Without Take the Lead, the family might ask for a cheaper treatment or decline the surgery. Figure 2 shows how the case study handles this visit instead.

Figure 2: Take the Lead Case Study Timeline



In the waiting room, one of the parents uses a phone to sign up as a Member and enters Biscuit's information and the $2,400 estimate. The hospital's front desk staff member opens the request on the hospital account and confirms the amount. Take the Lead sends a $500 Minimum Down Payment and a Promise to Pay for the remaining $1,900, and the surgery begins within the hour. The next morning, the family creates a Story with a photo of Biscuit and shares the link by text message and social media.

Over the next 30 days, 41 donors give a total of $2,600. On day 30 the Story closes. The first $500 repays the Minimum Down Payment, the next $1,900 pays the rest of the hospital bill, and the family keeps the remaining $200 for Biscuit's follow-up visit. The hospital is paid in full, the family has no debt, and Biscuit is home and healthy. In the prototype, every payment and donation in this scenario uses test data, so no real money changes hands.

The other four households in the beta group follow the same steps with different pets, bills, and donation totals. At least one Story should close short of its goal, so the case study can show that the unpaid balance stays the Member's responsibility and that the hospital is still paid. At the end of the 30 days, the participants complete a short survey about how easy each step was to use.

3.4 Full Capability and Future Expansion

The prototype shows the main process, but the full real-world product would add features that the prototype will only simulate. A payment processor would move real money between donors, Take the Lead, hospitals, and Members, and identity checks would confirm each Member before a payment guarantee is approved. Hospitals could connect Take the Lead to their practice-management software, so bills and estimates would be sent automatically instead of typed by hand. Members and donors would get text message and email updates as a Story moves toward its goal.

A later version could also offer a payment plan for any balance left when a Story closes and a longer Story for pets with ongoing conditions such as diabetes or cancer. A mobile app, Spanish-language pages, and pet insurance quotes inside the resource pages would make Take the Lead useful to more owners. Hospitals could also see reports on how many treatments Take the Lead helped them avoid declining.

3.5 Broader User and Customer Base

After the beta test, Take the Lead could serve pet owners anywhere in the United States who face a veterinary bill they cannot pay at once. Independent hospitals, corporate hospital groups, and emergency and specialty hospitals would all benefit from bills that are backed before treatment begins. Animal shelters and rescue groups could run Stories for animals that need surgery before adoption. Employers that offer pet benefits and pet insurance companies could partner with Take the Lead to cover the costs their plans leave out, and community foundations could match donations for pets in their area.

4 Glossary

Authentication: the process of checking that users are who they claim to be, usually through a sign-up and login.

Beta Test: a trial of a product by a small group of real users before the product is released to the public.

Crowdfunding: raising money through many small donations, usually collected online.

Framework: a set of prebuilt code and tools that developers use as a starting point for building software.

Major Functional Component Diagram (MFCD): a diagram showing the main parts of a product and how those parts connect.

Member: a pet owner with a Take the Lead account who can request a payment guarantee and create a Story.

Minimum Down Payment: part of a veterinary bill that Take the Lead pays up front and later recovers from Story donations.

Object-Relational Mapper (ORM): a tool that lets program code work with a database without hand-written database queries.

Practice-Management Software: software that a veterinary hospital uses to schedule visits, keep records, and send bills.

Promise to Pay: a guarantee from Take the Lead to a veterinary hospital that a bill will be paid within 30 days.

Prototype: an early working version of a product that demonstrates its main features with test data.

Story: a Member’s 30-day fundraising page for a pet.

Use Case: a step-by-step example of how a user works with a product to reach a goal.

5 References

Hrynowski, Z., & Lall, J. (2025, April 15). *52% of U.S. pet owners skipped or declined veterinary care.* Gallup. https://news.gallup.com/poll/659057/pet-owners-skipped-declined-veterinary-care.aspx

Kemp, A. (2026, January 19). *Veterinarians say cost is the main driver of declined care.* Gallup. https://news.gallup.com/poll/700115/veterinarians-say-cost-main-driver-declined-care.aspx